Arba Minch University

Arba Minch Institute of Technology

Faculty of Computing & Software Engineering

Final exam for Real-Time and Embedded System

Instructor: Yilkal B.

Date: 13/12/2023
Time Allowed: 1:30hr

Total Marks: 40%

Target Group: G3 CS, regular

Name: _____ Add ☐

IDNo: _____

Section: _____

Attention

- ☞ Make sure the exam paper contains **3 pages** excluding the cover page.
- ☞ The question paper contains **25 questions** in 4 **parts.**
- ☞ Answer the question according to the instructions given for each part.
- ☞ Write your **Name** and **ID No.** on each page of the question paper.
- ☞ Write your answer neatly on the space provided.
- ☞ Use of pencils and red pens are not allowed
- ☞ ***Use the back of the pages for part V.***
- ☞ ***Cheating in any form will be punished without prior notice.***

Good luck!!!

Do Not Turn on This Page Until You Are Told To Do So

Part I: Say, True if the statement is "Correct" or say False if the statement is "Wrong" (1 mark each)

1. An RTOS is an operating system that is designed to meet the requirements of real-time applications, which require quick and unpredictable responses to events.
2. One-time programmable ROM is a type of memory that can only be reprogrammed more than once.
3. Resource Reservation Protocols is a routing protocol and not depend on it.
4. Suspended is a state of task in RTOS that doesn't require CPU time.
5. If a system doesn't have timing constraints, it is not real-time.
6. Real-time embedded systems are often run in highly resource-unconstrained environments

Part II: Fill the blank space with the appropriate word (1 mark each)

1. _____ is used in resource reservation protocol for filtering the packets.
2. _____ is a re-use of an already developed and functioning application in a new real-time system without going back to the design and analysis phase.
3. _____ is a kernel object that one or more tasks can acquire or release for in the real-time operating system.
4. _____ is designed and manufactured for one specific application and doesn't allow you to reprogram or modify it after it is produced
5. _____ is a collection of computers connected via the high-speed communication network
6. _____ is used in distributed systems to achieve synchronization.

Part III: Choose the best answer from the given alternatives (1 mark each)

1. Which of the following is not the characteristic of a real-time operating System?
   A. Small footprint
   B. Indeterminism
   C. All of the above
   D. None of the above
2. Which of the following is not a feature of a real-time operating System?
   A. Can't work in unpredictable environment
   B. Response times may be unpredictable
   C. All of the above
   D. None of the above

3. At which side resource reservation protocol is done for making the necessary changes in the real-time communication strategy?

A. Receiver Side B. Sender Side C. All of the Above D. None of the above

4. Which of the following is not an example of a specific purpose microprocessor?

A. Application-specific integrated circuit C. All of the above

B. Application-specific standard product D. None of the above

5. Which of the following is odd?

A. Cyclic scheduling B. Clock-driven C. Pre-run time D. None of the above

6. The resource reservation protocol is a _____ designed to reserve resources across a network for quality of service.

A. Network Layer B. Application Layer C. Presentation Layer D. None of the Above

7. Which of the following is not true about traffic shaping?

A. Support implantation of traffic remarking C. Both A and B

B. Cause packet to delay D. None of the above

8. Which of the following is true about memory management units _____?

A. Increase the cost of the system

B. Increase the power consumption of the system

C. Increase the time required to complete an operation

D. All of the above except B

E. All of the above

9. Preemptive, priority-based scheduling guarantees _____

B. Hard real-time functionality D. Both A and B

C. Soft real-time functionality E. None of the above

10. which one of the following can be used to solve the priority inversion problem _____?

A. Priority inheritance protocol C. Both A&B

B. Priority inversion protocol D. None of the above

Part V: Answer the following questions accordingly (18 total marks)

1. List and discuss at least four advantages and drawbacks of concurrency in real-time systems [4pt.].
2. What is a task, a semaphore, priority inversion, context switch, and deadlock in an RTOS, explain each in detail with an example [5].
3. List and discuss the difference between multiprogramming and multitasking in a real-time system (3).
4. List and explain resource sharing protocols with examples used in real-time and embedded system [6].

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